TEMPERATE CONTINENTAL STEPPE CLIMATE

TEMPERATE GRASSLAND WORLD MAP → CONTINENTALITY MECHANISM → RAINFALL-MOISTURE PROFILE → HEMISPHERE ASYMMETRY → CHERNOZEM-SOIL CHAIN → GRANARY CONVERSION TIMELINE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g3 • explicit approval of this exact generation is required • all prior artifacts unchanged

00 → 01 → 02 → 03 → 04 → 05 → 06 → 07 → 08 → 09 → 10 → 11 → E

STAGE 00 — TEMPERATE GRASSLAND WORLD MAP

TEMPERATE GRASSLAND WORLD MAP | UKRAINE-KAZAKHSTAN BELT, CONTINENTAL INTERIOR | NORTH AMERICA | GREAT PLAINS, USA AND CANADA | SOUTH AMERICA | ARGENTINA AND URUGUAY, MARITIME FRINGE | SOUTH AFRICA | ELEVATED INTERIOR PLATEAU

EURASIA → steppes: Ukraine-Kazakhstan belt, continental interior

NORTH AMERICA → prairies: Great Plains, USA and Canada → SOUTH AMERICA → pampas: Argentina and Uruguay, maritime fringe → SOUTH AFRICA → veld: elevated interior plateau

SYSTEM / DEFINITION
- EURASIA → steppes: Ukraine-Kazakhstan belt, continental interior
- NORTH AMERICA → prairies: Great Plains, USA and Canada

PROCESS / MECHANISM
- SOUTH AMERICA → pampas: Argentina and Uruguay, maritime fringe
- SOUTH AFRICA → veld: elevated interior plateau

SPATIAL / INDIA PATTERN
- AUSTRALIA → downs: eastern interior grasslands
- EURASIA → steppes: Ukraine-Kazakhstan belt, continental interior

HAZARD / LIMIT / RESPONSE
- NORTH AMERICA → prairies: Great Plains, USA and Canada
- SOUTH AMERICA → pampas: Argentina and Uruguay, maritime fringe

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SOUTH AMERICA → pampas: Argentina and Uruguay, maritime fringe.

STAGE 01 — CONTINENTALITY MECHANISM

CONTINENTALITY MECHANISM | DISTANCE FROM SEA | WEAK MARITIME MODERATION | STRONG HEATING OF CONTINENTAL MASS | STRONG RADIATIVE COOLING | LARGE ANNUAL TEMPERATURE RANGE, ESPECIALLY IN EURASIA

DISTANCE FROM SEA → weak maritime moderation → SUMMER → strong heating of continental mass → hot → WINTER → strong radiative cooling → cold → RESULT → large annual temperature range, especially in Eurasia

DEFINITION / COMPONENTS
- DISTANCE FROM SEA → weak maritime moderation
- SUMMER → strong heating of continental mass → hot

TRIGGER / MECHANISM
- WINTER → strong radiative cooling → cold
- RESULT → large annual temperature range, especially in Eurasia

EFFECT / INTERVENTION
- DISTANCE FROM SEA → weak maritime moderation
- SUMMER → strong heating of continental mass → hot

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESULT → large annual temperature range, especially in Eurasia.

STAGE 02 — RAINFALL-MOISTURE PROFILE

RAINFALL-MOISTURE PROFILE | TOTAL ANNUAL | ABOUT 25-75 CM, LOW TO MODERATE | SEASONAL MAXIMUM | LATE SPRING AND EARLY SUMMER | DRY MARGIN | SHORTER GRASS, STEPPE-DESERT TRANSITION | WET MARGIN

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
TOTAL ANNUAL → about 25-75 cm, low to moderate	DRY MARGIN → shorter grass, steppe-desert transition	TOTAL ANNUAL → about 25-75 cm, low to moderate
SEASONAL MAXIMUM → late spring and early summer	WET MARGIN → taller grass, steppe-forest boundary	SEASONAL MAXIMUM → late spring and early summer

DEFINITION / COMPONENTS
- TOTAL ANNUAL → about 25-75 cm, low to moderate
- SEASONAL MAXIMUM → late spring and early summer

TRIGGER / MECHANISM
- DRY MARGIN → shorter grass, steppe-desert transition
- WET MARGIN → taller grass, steppe-forest boundary

EFFECT / INTERVENTION
- TOTAL ANNUAL → about 25-75 cm, low to moderate
- SEASONAL MAXIMUM → late spring and early summer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SEASONAL MAXIMUM → late spring and early summer.

STAGE 03 — HEMISPHERE ASYMMETRY

HEMISPHERE ASYMMETRY | STEPPES AND PRAIRIES | LARGE CONTINENTAL MASS, EXTREME RANGE | PAMPAS, VELD, DOWNS | NARROWER LANDMASS, MARITIME MODERATION | CONTINENTAL MASS DETERMINES WINTER SEVERITY | EXAM TRAP | DO NOT TREAT ALL FIVE AS CLIMATICALLY IDENTICAL

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
NORTHERN → steppes and prairies: large continental mass, extreme range	CAUSE → continental mass determines winter severity	NORTHERN → steppes and prairies: large continental mass, extreme range
SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation	EXAM TRAP → do not treat all five as climatically identical	SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation

DEFINITION / COMPONENTS
- NORTHERN → steppes and prairies: large continental mass, extreme range
- SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation

TRIGGER / MECHANISM
- CAUSE → continental mass determines winter severity
- EXAM TRAP → do not treat all five as climatically identical

EFFECT / INTERVENTION
- NORTHERN → steppes and prairies: large continental mass, extreme range
- SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXAM TRAP → do not treat all five as climatically identical.

STAGE 04 — CHERNOZEM-SOIL CHAIN

CHERNOZEM-SOIL CHAIN | GRASSLAND COVER | DEEP ROOT NETWORK DECAYS IN PLACE | LOW LEACHING | BASE-RICH CONDITIONS PRESERVED | HUMUS ACCUMULATES | DARK, DEEP, FERTILE SOIL PROFILE | CHERNOZEM OR BLACK EARTH, AMONG WORLD'S MOST FERTILE

GRASSLAND COVER → deep root network decays in place → LOW LEACHING → base-rich conditions preserved → HUMUS ACCUMULATES → dark, deep, fertile soil profile → RESULT → chernozem or black earth, among world's most fertile

DEFINITION / COMPONENTS
- GRASSLAND COVER → deep root network decays in place
- LOW LEACHING → base-rich conditions preserved

TRIGGER / MECHANISM
- HUMUS ACCUMULATES → dark, deep, fertile soil profile
- RESULT → chernozem or black earth, among world's most fertile

EFFECT / INTERVENTION
- GRASSLAND COVER → deep root network decays in place
- LOW LEACHING → base-rich conditions preserved

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESULT → chernozem or black earth, among world's most fertile.

STAGE 05 — GRANARY CONVERSION TIMELINE

GRANARY CONVERSION TIMELINE | NATURAL STATE | PASTORAL GRASSLAND, LOW-DENSITY USE | RAIL + OCEAN FREIGHT | ACCESS TO DISTANT INDUSTRIAL MARKETS | MECHANISATION + HYV | HIGH OUTPUT PER WORKER ON LEVEL PLAINS | WORLD'S WHEAT GRANARIES

NATURAL STATE → pastoral grassland, low-density use → RAIL + OCEAN FREIGHT → access to distant industrial markets → MECHANISATION + HYV → high output per worker on level plains → RESULT → world's wheat granaries: Ukraine, prairies, pampas

DEFINITION / COMPONENTS
- NATURAL STATE → pastoral grassland, low-density use
- RAIL + OCEAN FREIGHT → access to distant industrial markets

TRIGGER / MECHANISM
- MECHANISATION + HYV → high output per worker on level plains
- RESULT → world's wheat granaries: Ukraine, prairies, pampas

EFFECT / INTERVENTION
- NATURAL STATE → pastoral grassland, low-density use
- RAIL + OCEAN FREIGHT → access to distant industrial markets

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESULT → world's wheat granaries: Ukraine, prairies, pampas.

STAGE 06 — ANTI-DETERMINISM FIREWALL

ANTI-DETERMINISM FIREWALL | PHYSICAL BASE | FERTILE SOIL + LEVEL RELIEF + SUITABLE CLIMATE | CONVERSION AGENTS | RAIL, FREIGHT, MECHANISATION, MARKET DEMAND | WITHOUT ACCESS | SAME SOIL, NO GRANARY FOR CENTURIES | ENDOWMENT IS NECESSARY, ACCESS IS SUFFICIENT

PHYSICAL BASE → fertile soil + level relief + suitable climate

CONVERSION AGENTS → rail, freight, mechanisation, market demand → WITHOUT ACCESS → same soil, no granary for centuries → CONCLUSION → endowment is necessary, access is sufficient

DEFINITION / COMPONENTS
- PHYSICAL BASE → fertile soil + level relief + suitable climate
- CONVERSION AGENTS → rail, freight, mechanisation, market demand

TRIGGER / MECHANISM
- WITHOUT ACCESS → same soil, no granary for centuries
- CONCLUSION → endowment is necessary, access is sufficient

EFFECT / INTERVENTION
- PHYSICAL BASE → fertile soil + level relief + suitable climate
- CONVERSION AGENTS → rail, freight, mechanisation, market demand

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONCLUSION → endowment is necessary, access is sufficient.

STAGE 07 — ECOLOGICAL-COST CHAIN

ECOLOGICAL-COST CHAIN | SOD REMOVAL | BARE FINE SOIL EXPOSED TO WIND | DROUGHT YEAR | VEGETATION FAILS, DEFLATION PEAKS | ORGANIC DECLINE | REDUCED HUMUS, COMPACTION RISK | AQUIFER DRAWDOWN

SOD REMOVAL → bare fine soil exposed to wind → DROUGHT YEAR → vegetation fails, deflation peaks → ORGANIC DECLINE → reduced humus, compaction risk → AQUIFER DRAWDOWN → irrigation in semi-arid belt depletes groundwater

DEFINITION / COMPONENTS
- SOD REMOVAL → bare fine soil exposed to wind
- DROUGHT YEAR → vegetation fails, deflation peaks

TRIGGER / MECHANISM
- ORGANIC DECLINE → reduced humus, compaction risk
- AQUIFER DRAWDOWN → irrigation in semi-arid belt depletes groundwater

EFFECT / INTERVENTION
- SOD REMOVAL → bare fine soil exposed to wind
- DROUGHT YEAR → vegetation fails, deflation peaks

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AQUIFER DRAWDOWN → irrigation in semi-arid belt depletes groundwater.

STAGE 08 — INDIA WHEAT GRANARY MAP

INDIA WHEAT GRANARY MAP | PUNJAB-HARYANA-WESTERN UP | INDO-GANGETIC ALLUVIAL PLAIN | ALLUVIAL, NOT CHERNOZEM (FUNDAMENTAL DIFFERENCE) | SOWN OCT-DEC, HARVESTED MAR-APR | HYV SEEDS + IRRIGATION + FERTILISER + MSP PROCUREMENT

PUNJAB-HARYANA-WESTERN UP → Indo-Gangetic alluvial plain

SOIL → alluvial, not chernozem (fundamental difference) → SEASON → Rabi: sown Oct-Dec, harvested Mar-Apr → ENABLERS → HYV seeds + irrigation + fertiliser + MSP procurement

SYSTEM / DEFINITION
- PUNJAB-HARYANA-WESTERN UP → Indo-Gangetic alluvial plain
- SOIL → alluvial, not chernozem (fundamental difference)

PROCESS / MECHANISM
- SEASON → Rabi: sown Oct-Dec, harvested Mar-Apr
- ENABLERS → HYV seeds + irrigation + fertiliser + MSP procurement

SPATIAL / INDIA PATTERN
- PUNJAB-HARYANA-WESTERN UP → Indo-Gangetic alluvial plain
- SOIL → alluvial, not chernozem (fundamental difference)

HAZARD / LIMIT / RESPONSE
- SEASON → Rabi: sown Oct-Dec, harvested Mar-Apr
- ENABLERS → HYV seeds + irrigation + fertiliser + MSP procurement

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ENABLERS → HYV seeds + irrigation + fertiliser + MSP procurement.

STAGE 09 — GREEN REVOLUTION PACKAGE

GREEN REVOLUTION PACKAGE | HYV SEMI-DWARF WHEAT | RAISED YIELD POTENTIAL | CANAL + TUBE-WELL IRRIGATION | REDUCED RAINFALL DEPENDENCE | CHEMICAL FERTILISERS | SUPPORTED INTENSIVE CULTIVATION | ASSURED FLOOR PRICE AND CENTRAL POOL STOCKS

HYV SEMI-DWARF WHEAT → raised yield potential → CANAL + TUBE-WELL IRRIGATION → reduced rainfall dependence → CHEMICAL FERTILISERS → supported intensive cultivation → MSP/PROCUREMENT → assured floor price and central pool stocks

DEFINITION / COMPONENTS
- HYV SEMI-DWARF WHEAT → raised yield potential
- CANAL + TUBE-WELL IRRIGATION → reduced rainfall dependence

TRIGGER / MECHANISM
- CHEMICAL FERTILISERS → supported intensive cultivation
- MSP/PROCUREMENT → assured floor price and central pool stocks

EFFECT / INTERVENTION
- HYV SEMI-DWARF WHEAT → raised yield potential
- CANAL + TUBE-WELL IRRIGATION → reduced rainfall dependence

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MSP/PROCUREMENT → assured floor price and central pool stocks.

STAGE 10 — SAVANNA-STEPPE COMPARISON

SAVANNA-STEPPE COMPARISON | TROPICAL SAVANNA VERSUS TEMPERATE STEPPE/PRAIRIE | WET-AND-DRY SEASONAL VERSUS LOW-TOTAL SPRING MAXIMUM | LEACHED LATERISING VERSUS BASE-RICH CHERNOZEM/PRAIRIE | PASTORAL-SUBSISTENCE VERSUS MECHANISED-COMMERCIAL GRAIN

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
LATITUDE → tropical savanna versus temperate steppe/prairie	SOIL → leached laterising versus base-rich chernozem/prairie	LATITUDE → tropical savanna versus temperate steppe/prairie
RAINFALL → wet-and-dry seasonal versus low-total spring maximum	ECONOMY → pastoral-subsistence versus mechanised-commercial grain	RAINFALL → wet-and-dry seasonal versus low-total spring maximum

DEFINITION / COMPONENTS
- LATITUDE → tropical savanna versus temperate steppe/prairie
- RAINFALL → wet-and-dry seasonal versus low-total spring maximum

TRIGGER / MECHANISM
- SOIL → leached laterising versus base-rich chernozem/prairie
- ECONOMY → pastoral-subsistence versus mechanised-commercial grain

EFFECT / INTERVENTION
- LATITUDE → tropical savanna versus temperate steppe/prairie
- RAINFALL → wet-and-dry seasonal versus low-total spring maximum

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RAINFALL → wet-and-dry seasonal versus low-total spring maximum.

STAGE 11 — STEPPE ANSWER SPINE

STEPPE ANSWER SPINE | CONTINENTAL INTERIOR GRASSLAND WITH LARGE ANNUAL RANGE | FIVE REGIONAL NAMES ON A WORLD MAP | MOISTURE DEFICIT, CHERNOZEM, GRANARY CONVERSION | ACCESS CONVERTED ENDOWMENT INTO RESOURCE

DEFINE → continental interior grassland with large annual range → LOCATE → five regional names on a world map → EXPLAIN → moisture deficit, chernozem, granary conversion → QUALIFY → anti-determinism: access converted endowment into resource

DEFINITION / COMPONENTS
- DEFINE → continental interior grassland with large annual range
- LOCATE → five regional names on a world map

TRIGGER / MECHANISM
- EXPLAIN → moisture deficit, chernozem, granary conversion
- QUALIFY → anti-determinism: access converted endowment into resource

EFFECT / INTERVENTION
- DEFINE → continental interior grassland with large annual range
- LOCATE → five regional names on a world map

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFY → anti-determinism: access converted endowment into resource.

EXTRA — SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | INDIA'S WHEAT GRANARY | PUNJAB, HARYANA, WESTERN UP | INDO-GANGETIC PLAIN, | PUNJAB IS DESCRIBED AS A MAJOR FOOD BASKET/GRANARY IN | PUNJAB'S HIGH CULTIVATED SHARE AND CROPPING INTENSITY ARE SOURCE-ERA | HARYANA'S WHEAT BELT SPANS IRRIGATED EASTERN, CENTRAL AND WESTERN | WHEAT IS A RABI CROP.

OPTIONAL PROCESS DEPTH
- India's wheat granary = Punjab, Haryana, western UP / Indo-Gangetic plain.
- Punjab is described as a major food basket/granary in the source text.
- Punjab's high cultivated share and cropping intensity are source-era indicators, not current statistics.
- Haryana's wheat belt spans irrigated eastern, central and western plains; use current district maps.

DATA / INSTITUTION LIMIT
- Wheat is a Rabi crop: sown Oct-Dec, harvested Mar-Apr in source notes.
- Green Revolution package: HYV + irrigation + fertiliser + MSP/procurement.
- Soil is alluvial , not steppe chernozem.
- The source-era text records a substantial marketed surplus; current share requires a dated source.

CONTEMPORARY PARALLEL
- Surplus states: Punjab, Haryana, UP, Rajasthan, MP.
- Deficit destinations include Maharashtra, West Bengal, Bihar and Delhi.
- NOT: India's wheat belt sits on chernozem like the steppe → it sits on Indo-Gangetic alluvial soil.
- NOT: Wheat is a Kharif monsoon crop → Wheat is a Rabi winter crop.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels